

WENYU YANG

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EDUCATION

Dartmouth College

Sep. 2025 – Nov. 2025

Engineering Exchange Program

CGPA 3.89 / 4.0

- Relevant Coursework: Math for Machine Learning, Artificial Intelligence, Distributed Computing

The Chinese University of Hong Kong

Sep. 2023 – Jul. 2027

Bachelor of Artificial Intelligence: Systems and Technologies, Dean's List

CGPA 3.65 / 4.0

- Relevant Coursework: Introduction to Data Science, Human-Computer Interaction, Introduction to Computing Using C++, Statistics for Engineers, Data Structures

PROFESSIONAL EXPERIENCE

HealthHope (Beijing) Technology Co., Ltd.

May 2025 – Jul. 2025

Back-end Engineer (Python Data Engineering & APIs)

Beijing, China

- Developed an concurrent, automated Pandas-based ETL data cleaning pipeline using Queue and Threading, with null value imputation, directory-level deduplication, and outlier removal
- Intercepted undisclosed backend APIs through reverse engineering to directly extract nested JSON payloads, bypassing DOM rendering and significantly reducing computational overhead to the microsecond level
- Handled complex dynamic rendering and low-level anti-scraping mechanisms, integrated undetected_chromedriver to bypass Cloudflare's stringent protection and CAPTCHA verification
- Automated the post-processing of large volumes of unstructured text into highly structured JSON/CSV formats, providing reliable inputs for downstream data analysis and machine learning model training

ACADEMIC EXPERIENCE

Minds, Machines, and Society Group, Dartmouth College

Sep. 2025 – Present

Large Language Model (LLM) Application & Optimization

Hanover, NH

- Engineered pre-training pipeline using Hugging Face Accelerate for distributed training, and designed controlled ablation studies to evaluate model cross-lingual representation learning and downstream classification generalisation
- Overcame stringent CUDA Out-of-Memory (OOM) constraints during pre-training by combining mixed-precision training (FP16) and gradient accumulation, simulating an Effective Batch Size of 2048
- Trained a custom SentencePiece Unigram tokeniser and evaluated mitigation trade-offs on different solutions to resolve Out-of-Vocabulary (OOV) bottlenecks in the Ancient Chinese tokeniser
- Architected a TensorBoard monitoring suite to track convergence curves and gradient norms, systematically debugging training loss plateaus through rigorous learning rate and hyperparameter tuning

Department of Systems Engineering, CUHK

Jan. 2024 – May 2024

Machine Learning for Obesity Risk Prediction

Hong Kong, China

- Engineered a predictive classification model utilising RandomTree algorithms to analyse high-dimensional lifestyle and demographic datasets, successfully achieving 75.2% validation accuracy
- Conducted rigorous exploratory data analysis (EDA) and statistical visualisation via Tableau and Matplotlib to isolate critical health risk factors and feature correlations

LEADERSHIP EXPERIENCE

uPals Senior Wellness Ambassador & Team Leader

Dec. 2025 – Present

Wellness And Counselling Centre, CUHK

Hong Kong

- Certified in leadership and crisis intervention; led a team of 30+ peers in executing large-scale support initiatives

SKILLS

Languages & Tools: Python, C/C++, SQL, HTML, JSON, Git, Linux, VS Code

AI & Data Science: OpenCV, PyTorch, Hugging Face, Pandas, Tableau, Matplotlib, LLM Fine-tuning

Data Eng & Web: ETL Pipelines, Multithreading, Selenium, BeautifulSoup, REST APIs